

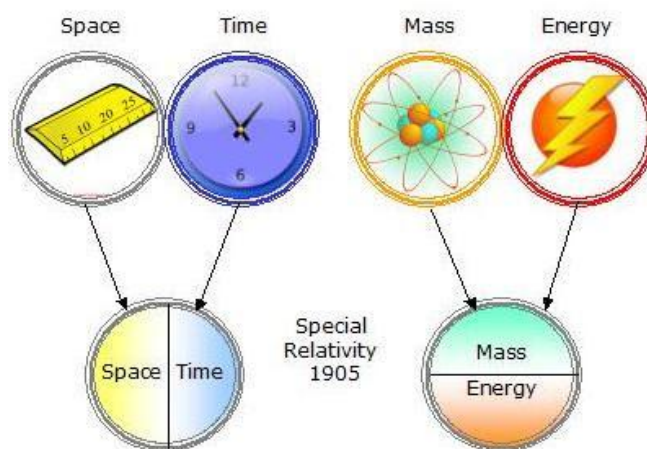
## 22PH102 – ELECTROMAGNETISM AND MODERN PHYSICS

### Unit-IV: MODERN PHYSICS\_LM4

#### Consequences of the Special Theory of Relativity

❖ Relativistic mass variation

❖ Relativistic Energy

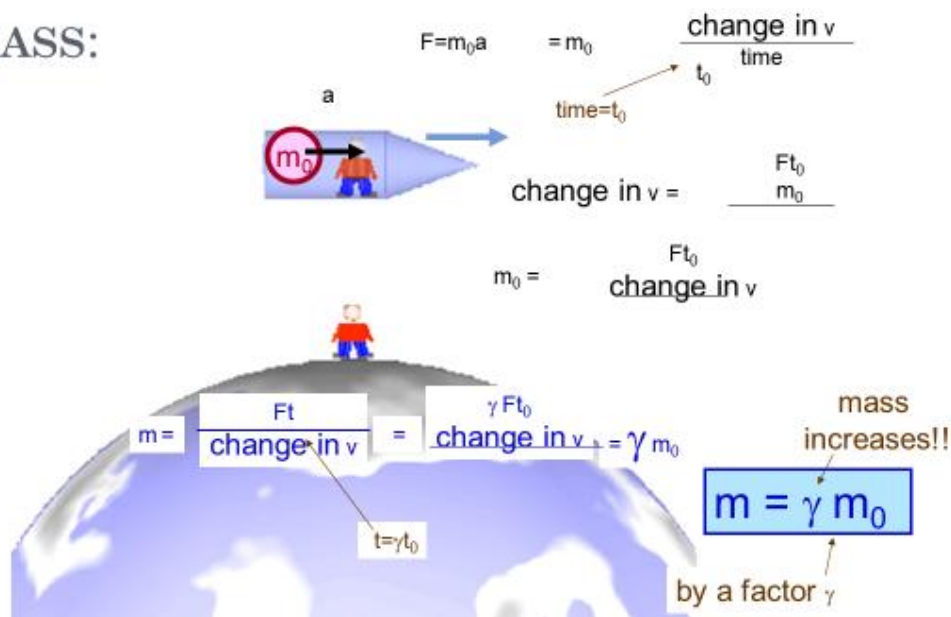


In Einstein's Special Theory of Relativity, the concept of mass undergoes a transformation at relativistic speeds. This phenomenon, known as relativistic mass variation, describes how the mass of an object changes as its velocity approaches the speed of light. In this lecture material, we will explore the theory behind relativistic mass variation, discuss its implications, and provide examples to illustrate this concept.

## Mass in Relativity

- In older treatments of relativity, conservation of momentum was maintained by using “relativistic mass”.
- Today, mass is considered to be **invariant**.
  - That means it is **independent of speed**.
- The mass of an object in all frames is considered to be the mass as measured by an observer at rest with respect to the object.

MASS:



## RELATIVISTIC MASS INCREASE

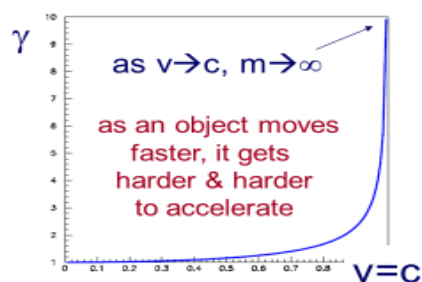
$m_0$  = mass of an object when it is at rest

→ “rest mass”

mass of a moving object increases

$$m = \gamma m_0$$

by the  $\gamma$  factor





## Relativistic Momentum

### Relativistic Momentum

$$p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Note: for  $v \ll c$   $p = mv$ Note: for  $v = c$   $p = \text{infinity}$ 

### Relativistic Energy

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Note: for  $v = 0$   $E = mc^2$ Note: for  $v \ll c$   $E = mc^2 + \frac{1}{2}mv^2$ Note: for  $v = c$   $E = \text{infinity}$  (if  $m$  is not 0)Objects with mass always have  $v < c$ !

- To account for conservation of momentum in all inertial frames, the definition must be modified.

$$p \equiv \frac{mv}{\sqrt{1 - v^2/c^2}} = \gamma mv$$

- $v$  is the speed of the particle;  $m$  is its mass as measured by an observer at rest with respect to the mass.
- When  $v \ll c$ , the denominator approaches 1 and so  $p$  approaches  $mv$ .

### Relativistic Energy:

The famous Einstein's relation for energy is

$$E = mc^2 \quad (\text{This includes both the kinetic energy and rest mass energy for a particle})$$

$$KE = mc^2 - m_0c^2 \quad E = mc^2 = E_0 + KE = m_0c^2 + KE$$

If the object is moving, then the relativistic energy is

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Mass can be defined as Energy  
Energy can be defined as Mass

The relativistic energy expression is the tool used to calculate the binding energies of nuclei and the energy yield of nuclear fission and fusion

From  $E = mc^2$  to the atomic bomb

- The definition of kinetic energy requires modification in relativistic mechanics.

- $KE = \gamma mc^2 - mc^2$

- The term  $mc^2$  is called the *rest energy* of the object and is independent of its speed.
- The term  $\gamma mc^2$  is the *total energy*,  $E$ , of the object and depends on its speed and its rest energy.

### Energy and Relativistic Momentum

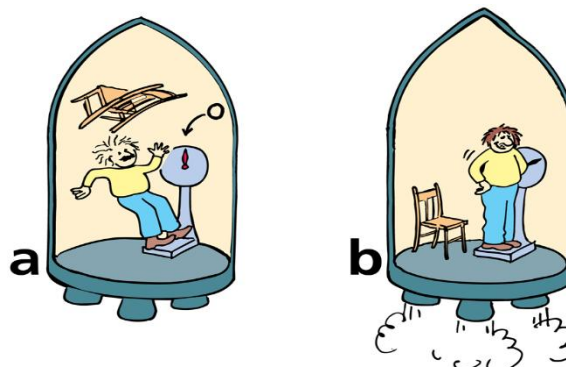
- It is useful to have an expression relating total energy,  $E$ , to the relativistic momentum,  $p$

- $E^2 = p^2 c^2 + (mc^2)^2$

- When the particle is at rest,  $p = 0$  and  $E = E_R = mc^2$
- Massless particles ( $m = 0$ ) have  $E = pc$
- This is also used to express masses in energy units.
  - Mass of an electron =  $9.11 \times 10^{-31}$  kg gives a rest energy of 0.511 MeV
  - Conversion:  $1 \text{ u} = 931.494 \text{ MeV}/c^2$

### **General Relativity Gravity & Acceleration**

- Everything inside is weightless when the spaceship isn't accelerating.
- When the spaceship accelerates, an occupant inside feels "gravity."



## SPECIAL RELATIVITY

### Theory of Relativity in Real Life

Time dilation

$$t' = t \times \sqrt{1 - \frac{v^2}{c^2}}$$

Length contraction

$$l' = \frac{l}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Mass dilation

$$m' = m \times \sqrt{1 - \frac{v^2}{c^2}}$$



Electromagnets



Global Positioning System



Yellow Color of Gold



No corrosion in Gold



Mercury is Liquid



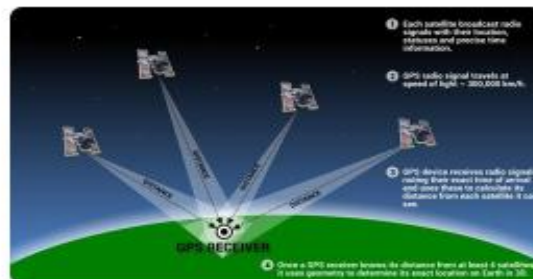
Your Old TV

### Applications of Special Theory of Relativity

Atom Bomb



Global Positioning System (GPS)



## Examples of Relativistic Mass Variation:

### Example 1: Particle Accelerators

Particle accelerators, such as the Large Hadron Collider (LHC), accelerate subatomic particles to velocities close to the speed of light. As particles gain



kinetic energy and approach relativistic speeds, their relativistic mass increases.

This increase in mass enables scientists to study particle collisions at higher energies, providing insights into the fundamental constituents of matter and the laws of physics.

### **Example 2: Nuclear Fusion in Stars**

In the cores of stars, nuclear fusion processes occur where hydrogen nuclei (protons) combine to form helium nuclei. These fusion reactions release enormous amounts of energy. According to Einstein's mass-energy equivalence principle, the increase in binding energy during fusion corresponds to a decrease in the total mass of the system. As particles gain kinetic energy in these extreme conditions, their relativistic mass increases, contributing to the overall energy balance of the fusion process.

### **Example 3: Cosmic Ray Particles**

Cosmic rays are high-energy particles originating from outer space. When these particles interact with Earth's atmosphere, they produce secondary particles that travel at relativistic speeds. As these particles gain kinetic energy, their relativistic mass increases. This increase in mass affects the observed characteristics and behaviors of cosmic ray showers detected by ground-based observatories, providing valuable information about the origins and properties of cosmic rays.

**Example 4: Particle Decay**

In particle physics, subatomic particles are often produced in high-energy collisions and subsequently decay into lighter particles. The relativistic mass of these particles plays a crucial role in determining their decay processes and lifetimes. As particles approach the speed of light, their relativistic mass increases, affecting the probability and kinematics of particle decay processes.